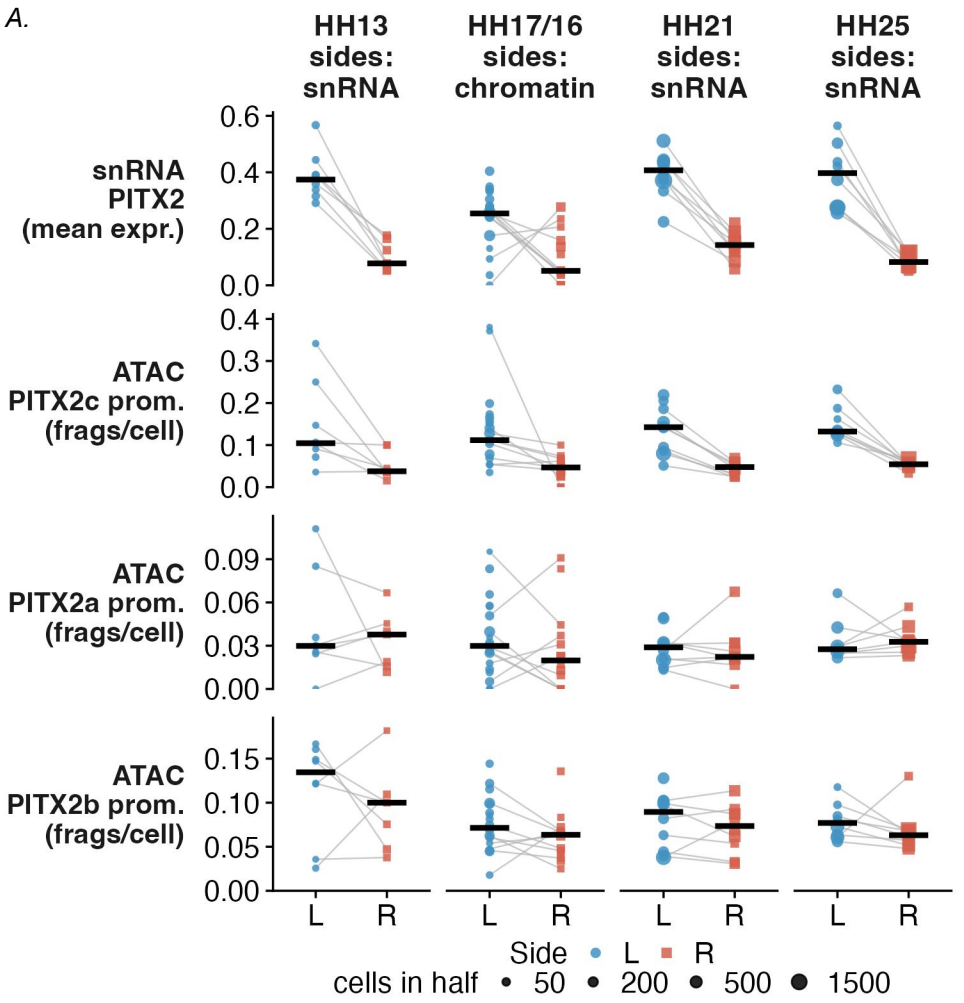
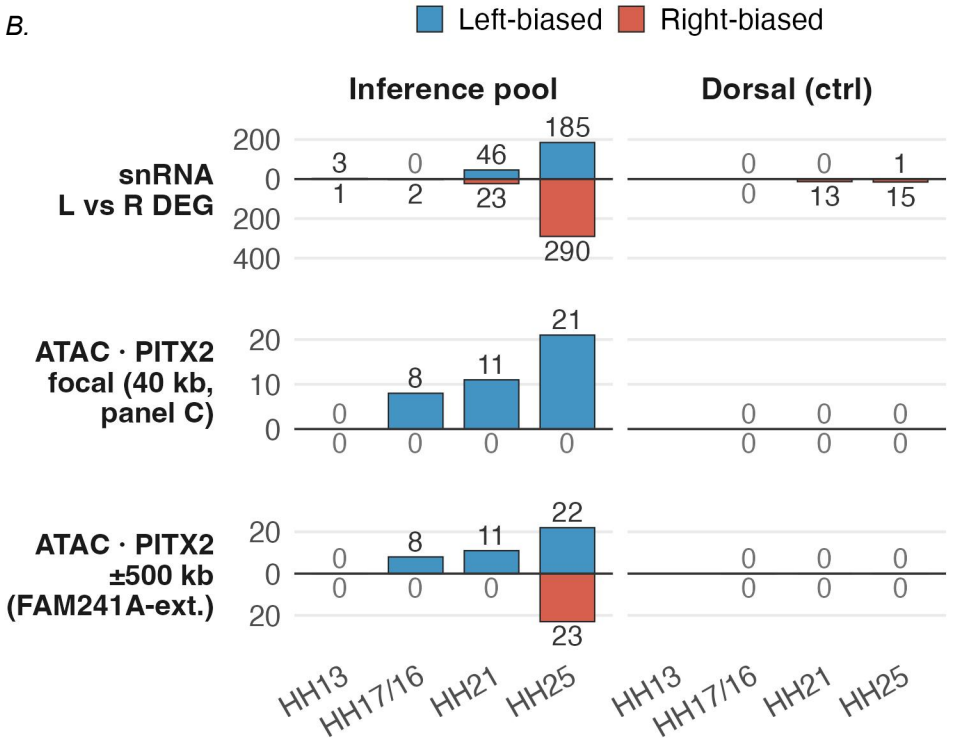
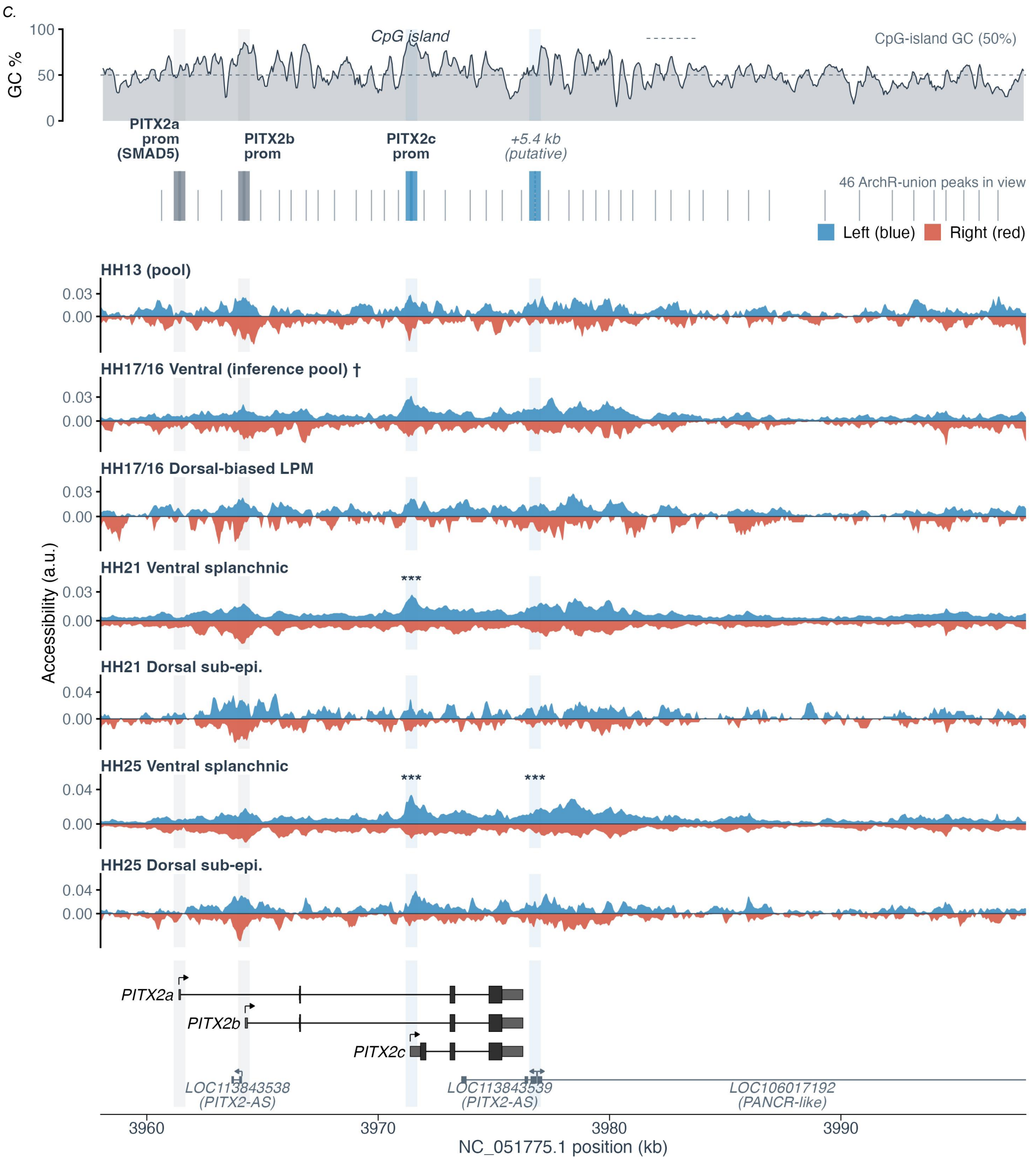
A.


HH13 sides: snRNA
HH17/16 sides: chromatin
HH21 sides: snRNA
HH25 sides: snRNA
snRNA PITX2 (mean expr.)
ATAC PITX2c prom. (frags/cell)
ATAC PITX2a prom. (frags/cell)
ATAC PITX2b prom. (frags/cell)
L
R
Side L R
cells in half 50 200 500 1500

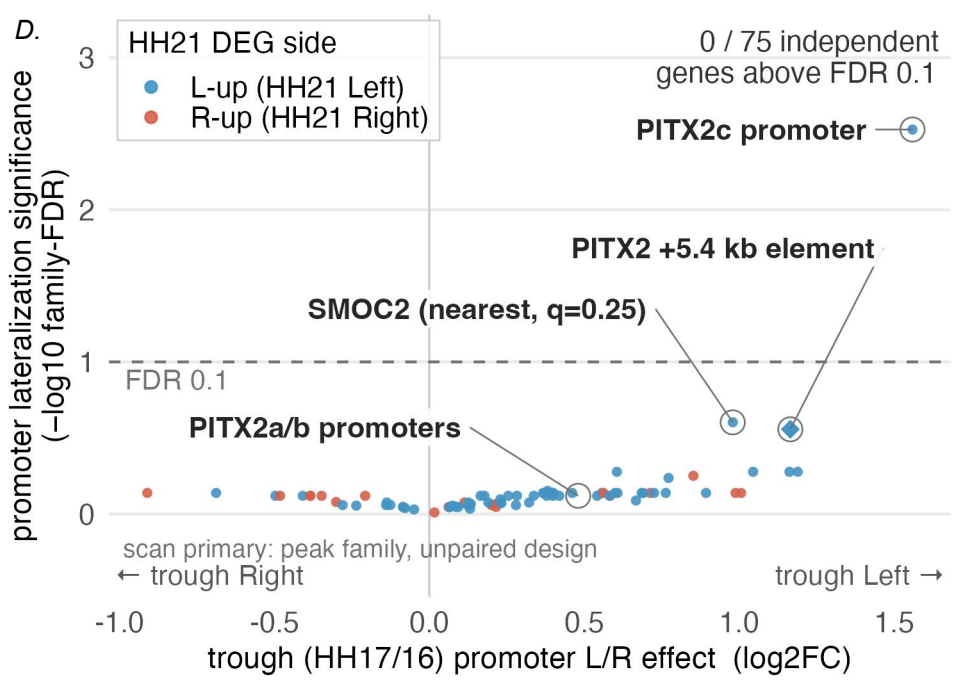
B.


Left-biased
Right-biased
Inference pool
Dorsal (ctrl)
snRNA L vs R DEG
ATAC · PITX2 focal (40 kb, panel C)
ATAC · PITX2 ±500 kb (FAM241A-ext.)
HH13
HH17/16
HH21
HH25

C.


GC %
CpG island
CpG-island GC (50%)
PITX2a prom (SMAD5)
PITX2b prom
PITX2c prom
+5.4 kb (putative)
PITX2a
46 ArchR-union peaks in view
Left (blue)
Right (red)
HH13 (pool)
HH17/16 Ventral (inference pool) †
HH17/16 Dorsal-biased LPM
HH21 Ventral splanchnic
HH21 Dorsal sub-epi.
HH25 Ventral splanchnic
HH25 Dorsal sub-epi.
Accessibility (a.u.)
***
PITX2a
PITX2b
PITX2c
LOC113843538 (PITX2-AS)
LOC113843539 (PITX2-AS)
LOC106017192 (PANCR-like)
NC_051775.1 position (kb)

D.


HH21 DEG side
L-up (HH21 Left)
R-up (HH21 Right)
0 / 75 independent genes above FDR 0.1
PITX2c promoter
PITX2 +5.4 kb element
SMOC2 (nearest, q=0.25)
FDR 0.1
PITX2a/b promoters
promoter lateralization significance (−log10 family-FDR)
scan primary: peak family, unpaired design
← trough Right
trough Left →
trough (HH17/16) promoter L/R effect (log2FC)